

A PROJECT TITLE

ON

Fraudulent Transaction Clustering

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

In

C.S.I.T

A.Y.: 2026-27

by

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C.S.I.T



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DEPARTMENT OF _____

Name of Title: **Fraudulent Transaction Clustering**

1. Abstract:

Financial fraud is a critical challenge in the digital economy, characterized by constantly evolving tactics that often bypass traditional rule-based and supervised detection systems. This project explores the application of unsupervised machine learning, specifically clustering algorithms, to identify fraudulent transactions without the need for extensively labeled datasets. By grouping transactions based on behavioral and transactional similarities, the model establishes a baseline of normal activity. Outliers and transactions that fall outside these dense clusters are subsequently flagged as potential fraud. This approach offers a dynamic and scalable solution capable of adapting to novel, previously unseen fraud patterns, ultimately enhancing the security and anomaly detection capabilities of modern financial systems.

2. INTRODUCTION:

As digital payment systems and online banking expand, financial institutions face an unprecedented surge in transaction volumes, accompanied by a proportional increase in fraudulent activities. Fraudsters continuously adapt their methods, rendering traditional, static rule-based systems ineffective and prone to generating overwhelming numbers of false positives. While supervised machine learning models have improved detection rates, they heavily rely on historically labeled data. This dependency makes them highly vulnerable to "zero-day" or entirely new fraud techniques that have not yet been categorized.

To address this critical limitation, this project proposes an unsupervised machine learning approach leveraging clustering techniques. Unlike supervised models, clustering algorithms—such as K-Means, DBSCAN, or Hierarchical Clustering—do not require pre-labeled data to function. Instead, they analyze the inherent structure of the dataset, grouping transactions based on shared features such as transaction amount, geographical location, time, and frequency.

By mathematically defining what constitutes "normal" customer behavior through these clusters, any anomalous transactions that deviate significantly from established patterns can be isolated. This study aims to implement, visualize, and evaluate these clustering models to demonstrate their efficacy in detecting complex, evolving financial fraud in a highly imbalanced data environment.

3. SOFTWARE REQUIREMENTS:

Programming Language: Python 3.8 or higher.

Development Environment: Jupyter Notebook or Google Colab (for interactive data exploration and visualization), and VS Code or PyCharm (for final script structuring).

Data Manipulation & Analysis:

- Pandas (for data cleaning and DataFrame manipulation)
- NumPy (for numerical and matrix operations)

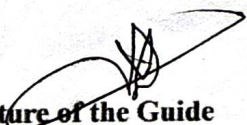
Machine Learning Libraries:

- Scikit-learn (provides the core algorithms like K-Means, DBSCAN, and evaluation metrics like Silhouette Score).

Data Visualization:

- Matplotlib and Seaborn (for plotting static cluster distributions and correlation matrices)
- Plotly or PCA/t-SNE tools (for interactive 2D/3D visualization of high-dimensional transaction data).

Version Control: Git (for tracking code changes) and GitHub/GitLab (for repository hosting).



Signature of the Guide



Course Instructor